

Ugo Giordano

Senior Software Engineer, Ph.D.

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SUMMARY

Senior Software Engineer at Red Hat, contributing to Kubernetes operator source code and building cloud-native security tooling for an enterprise AI platform. Engineers Go libraries, admission webhooks, and AI-powered DevSecOps automation to harden 22 platform operators. PhD in Software Reliability Engineering from University of Naples Federico II. Previously at Nokia Bell Labs researching Software-Defined Networking (SDN) control plane reliability.

EXPERIENCE

Senior Software Engineer, Platform Team for OpenShift AI

Nov 2024 – Present

Red Hat

Remote

- Contributed 50+ merged PRs to the platform's core Kubernetes operator codebase, spanning component lifecycle, security hardening, and coordination across 20+ repositories
- Identified and remediated a platform-wide RBAC misconfiguration across 22 operators by engineering a Go library enforcing namespace-scoped Roles over cluster-wide ClusterRoles, validated across 200 trials on two production clusters with zero performance regression
- Architected PipeClear, a two-layer ML pipeline validation system (Python KFP plugin + Go admission webhook) blocking untrusted code execution in Kubeflow Pipelines; presented at the KFP Community Call
- Built an architecture analyzer in Go that extracts CRDs, RBAC rules, security contexts, and network policies from operator source code, enabling automated cross-component security audits
- Automated a CI/CD-integrated readiness assessment across 135+ repositories, driving +10 to +28 point readiness-score improvements in 4 target repos through AI-powered analysis and targeted remediation

Software Maintenance Engineer

Aug 2021 – Oct 2024

Red Hat

Remote

- Resolved critical production escalations for enterprise OpenShift clients in financial services and telecom, diagnosing multi-component failures across CRI-O, etcd, SDN/OVN, and CSI drivers under strict SLAs

Freelance Software Engineer & Architect (part-time)

Jan 2025 – Present

Independent consultancy

Remote

- Designed and delivered healthcare and public administration systems, including ROC (Rete Oncologica Campania), from architecture through production deployment

Freelance Software Engineer & Architect (part-time)

Feb 2021 – Dec 2023

Independent consultancy (KIRANET Srl, Software Care Srl)

Remote

- Designed and shipped an integrated chronic disease management system for public healthcare, owning architecture, full-stack development, and technical advisory

Project Manager, Senior Software Engineer

Oct 2017 – Jul 2021

RisLab Srl

Naples

- Led design and delivery of a virtualized cyber-physical protection system for critical infrastructure, correlating PSIM and SIEM events to cut mean-time-to-detect for physical-cyber threats

PhD Intern, Member of Technical Staff

Apr 2016 – May 2017

Nokia Bell Labs

New Jersey, USA

- Built a cloud benchmarking framework for NetUNIX (carrier-grade Network OS), enabling in-production performance testing of SDN control planes for telecom operators

EDUCATION

Ph.D. in Electrical Engineering and Information Technology

2014 – 2017

University of Naples Federico II

Naples

M.Sc. in Computer Engineering, 110/110 cum laude ($\approx 3.94/4.0$ GPA)

2010 – 2013

University of Naples Federico II

Naples

B.Sc. in Computer Engineering, 103/110 ($\approx 3.40/4.0$ GPA)

2006 – 2010

University of Naples Federico II

Naples

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, JavaScript, Java, C/C++, Bash, SQL
Security: RBAC, Admission Webhooks, Semgrep, Gitleaks, tree-sitter, ast-grep
Platforms: Kubernetes, OpenShift, Docker/Podman, GitHub Actions, CI/CD
Frameworks: controller-runtime, FastAPI, React, PatternFly, Astro
Data & Tools: Protocol Buffers, PostgreSQL, SQLite, Git

CERTIFICATIONS

Red Hat Certified Architect in Infrastructure (RHCA) | Red Hat Certified Engineer (RHCE) | RHCSA
RH Specialist in OCP Data Foundation | RH Specialist in OCP Virtualization | RH Specialist in OCP Automation
RH Certified OCP Application Developer | RH Specialist in Containers

PUBLICATIONS

In Production Performance Testing of SDN Control Plane for Telecom Operators	DSN 2018
Prioritizing Correction of Static Analysis Infringements for Cost-Effective Sanitization	SER&IP 2015
Facing Cyber-Physical Security Threats by PSIM-SIEM Integration	EDCC 2019

HONOURS

Certificate of Appreciation Nokia Bell Labs, New Jersey, USA	2017
International Olympiad in Informatics (IOI) Kenosha, Wisconsin, USA	2003